



RAJALAKSHMI ENGINEERING COLLEGE

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

EVENTURA AI

An Agentic AI Event Planning and Coordination Platform

AD23731 | AGENTIC AI | FINAL PROJECT REVIEW

SUBMITTED BY



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Year/Sem: IV / VIII

FACULTY GUIDE



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Project Overview



AIM

Build an intelligent AI system for automated event planning and coordination.

Deliver personalized, constraint-aware event plans.



From a request
→ to a complete plan



APPROACH

Use multi-agent orchestration with Agentic RAG.

Combine retrieval, reasoning, validation, and human approval.



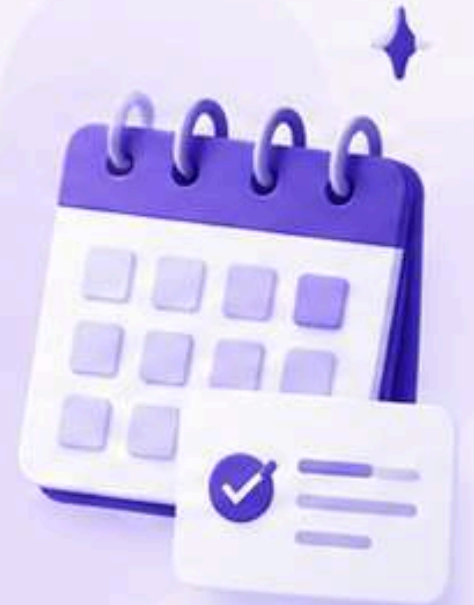
Coordinated agents
instead of a single chatbot



MVP

Plan events, select vendors, and manage budgets.

Support validation, replanning, and mock booking.



End-to-end planning
with human approval

The Problem

Event planning is a **coordination** problem, not a search problem.

A REAL EVENT SCENARIO

A 2-day wedding in Chennai



TODAY'S TOOLS ARE FRAGMENTED



Search tools

Google, wedding portals



Find vendors



Chatbots

General AI assistants



Generate suggestions



Spreadsheets

Manual tracking



Track budget



But nobody coordinates the decisions.



One vendor choice changes **every other decision.**



Meet Eventura AI

Your event planner, powered by a team of **AI agents**.



User says:

“Plan a 2-day wedding in Chennai for 500 guests within ₹10.2L.”

Eventura

New Event

My Events

Vendors

Marketplace

Evaluations

Create a New Event

Tell us about your event and let our AI agents create a personalized, comprehensive plan.

1

Event Details

2

Preferences

3

Review

4

Generate Plan

What type of event are you planning?

Wedding

Birthday

College Fest

Corporate Event

Describe your event

Plan a 2-day wedding in Chennai for 500 guests within ₹10.2L.

54/500

Event Date

20 December 2023

Guest Count

500

Location

Chennai

Budget (INR)

₹ 10,20,000

Next Step →

From your vision
to a beautifully planned event.

Eventura **does**:



UNDERSTAND

Extract requirements

Convert your request into structured event details, constraints and preferences.



CHECK

Test feasibility

Validate venue availability, budget feasibility and constraint satisfaction.



RESEARCH

Retrieve vendors + evidence

Find relevant vendors using agentic RAG (SQL + vector search) with real grounded information.



OPTIMIZE

Budget + negotiation

Build a balanced plan within budget and propose the best vendor mix.



CRITIQUE

Detect conflicts

Self-critique the plan for missing categories, constraint violations and better alternatives.



APPROVE

Human-in-the-loop

You review, modify or approve before any mock booking.



A coordinated event plan —
not a chatbot response.

Objectives & Scope



Objectives



Convert natural-language requests into structured requirements



Check feasibility, retrieve vendor and knowledge evidence



Reason about budget, negotiate and detect conflicts



Critique plans and replan through a reflection loop



Require human approval before mock booking



Expose agent activity for transparency



Compare No-RAG, Basic-RAG and Agentic-RAG with measurable metrics



In scope

Weddings, birthdays and college fests, a synthetic vendor marketplace (~95 vendors), mock booking, React dashboard with review and disruption simulation.



Out of scope

Real vendor APIs, real payments and bookings, real market data, and live availability. The system is a prototype, not a production booking platform.



Users & Stakeholders



Individual organizers

Plan weddings, birthdays and family functions in natural language.



Families

Coordinate celebrations with shared preferences and approvals.



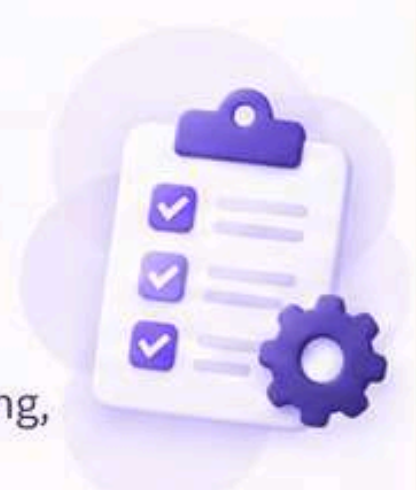
College clubs

Plan fests within tight budgets across several vendors.



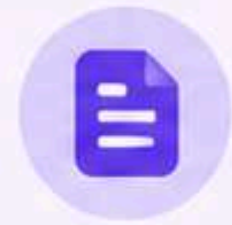
Professional organizers

Preliminary planning, vendor discovery, budget analysis and monitoring.



Other stakeholders

- Vendors (service providers)
- Event managers
- System administrators
- The AI system as a decision-support component



Key user requirements

- Natural-language input
- Vendor and budget views
- Evidence inspection
- Agent activity
- Approve/reject
- Disruption simulation
- Transparency, traceability, controlled autonomy



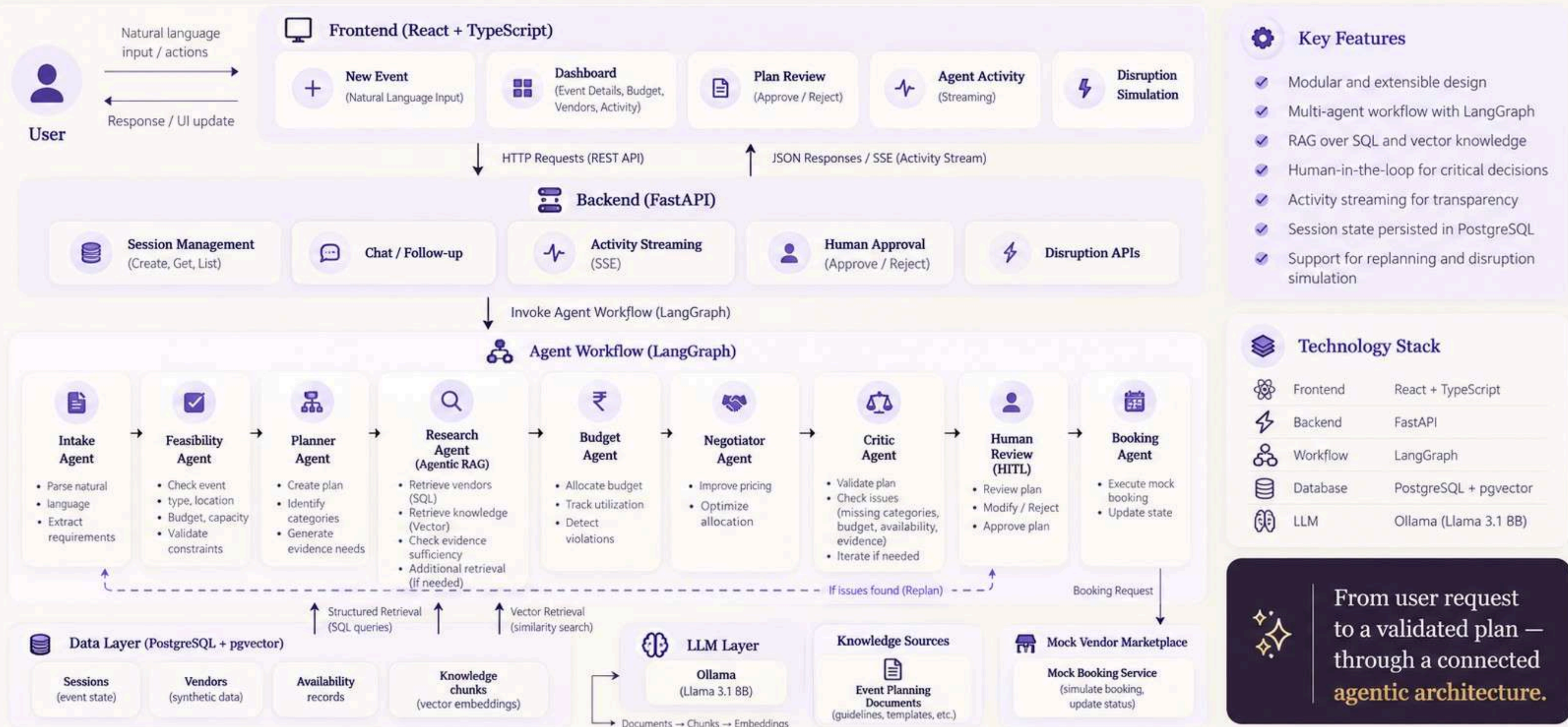
Proposed Agentic AI Solution





System Architecture

A modular, end-to-end architecture for **agentic event planning**.





The Agentic Brain

A team of specialized agents that **plan**, **validate** and **optimize** together.



Shared State
(PostgreSQL + LangGraph)

- Event requirements
- Selected vendors
- Budget and allocation
- Retrieved evidence
- Critic feedback
- Rejected vendors
- Booking status



Reflection Loop

Generate → Critique → Identify issues → Replan → Critique again



Human-in-the-Loop

Final review and approval before any booking.

8 Specialized Agents



Intake

Natural language to structured requirements



Feasibility

Budget, location, capacity and availability checks



Planner

Creates plan, sets evidence needs, replans



Research

Agentic RAG over SQL and vectors



Budget

Allocation, utilization and constraint checks



Negotiator

Improves pricing and proposes better options



Critic

Validates plan and detects conflicts



Booking

Mock booking after approval

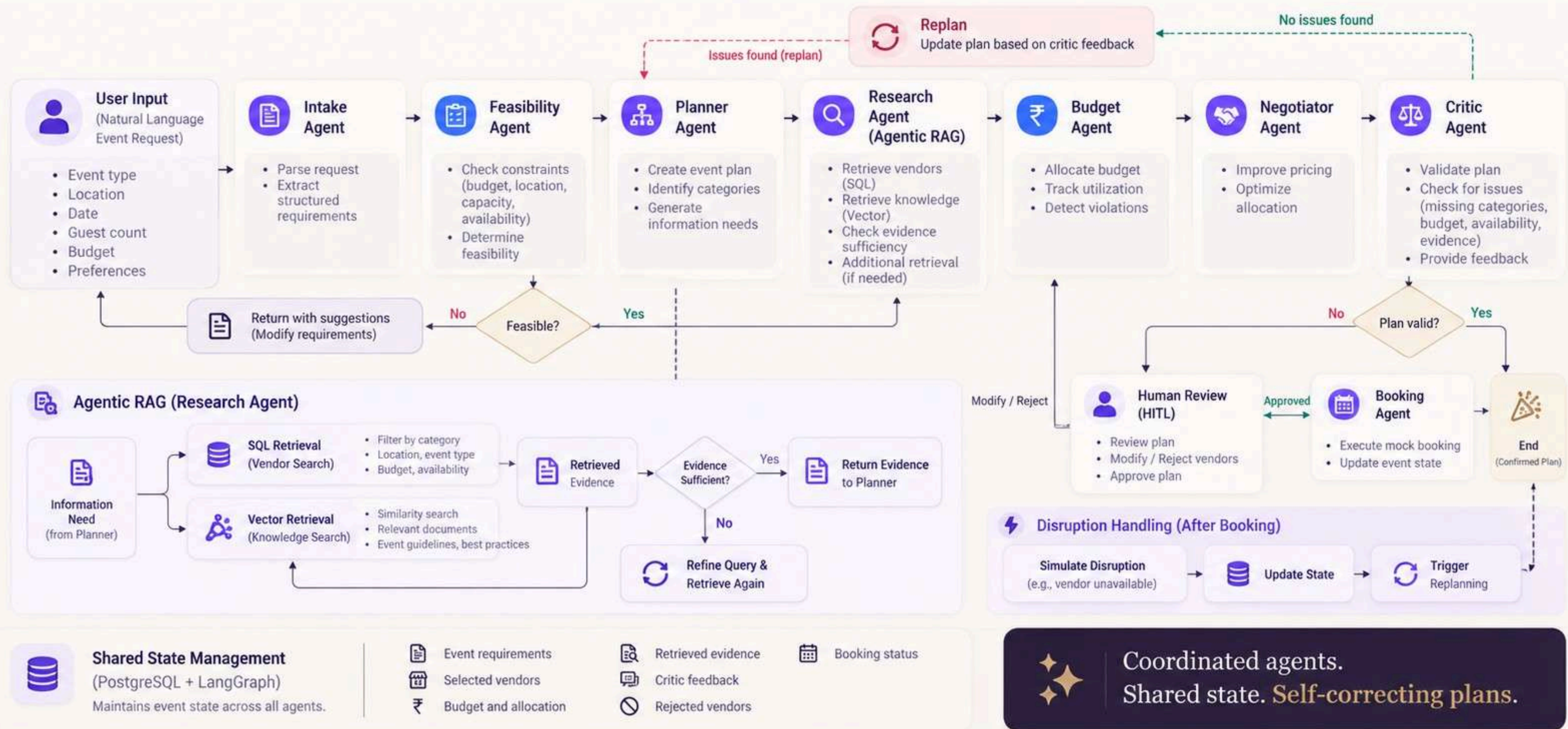


One shared state.

A coordinated plan. **Real validation.**

Agent Workflow & Orchestration

From user request to a validated event plan through a **multi-agent workflow**.



State, Reflection & Human-in-the-Loop

A shared state, a **self-correcting loop** and **human control** at key decisions.



Shared Structured State

All agents read and write to a single state, persisted in PostgreSQL (LangGraph).

	Event requirements	event_type, location, date, guest_count, budget, preferences
	Feasibility details	constraints, availability, feasibility results
	Plan & selected vendors	vendor_ids, categories, proposed plan
	Budget & allocation	total_budget, allocation, utilization, violations
	Retrieved evidence	SQL results, vector knowledge, source references
	Negotiation log	pricing options, alternatives
	Critic feedback	issues, missing categories, constraint violations
	Rejected vendors	rejected_vendor_ids, reasons
	Booking status	booking actions, status, update history



Shared structured state

Agents read and write one state: requirements, feasibility, plan, evidence, retrieval_rounds, budget, negotiation_log, critic_feedback, rejected_vendor_ids and bookings. A PostgreSQL-backed LangGraph checkpoint lets the run pause and resume.



Reflection loop

Generate, critique, identify problems, replan, critique again. The Critic checks budget, missing categories, evidence and availability. Rejected vendor IDs carry through replanning so they are not chosen again.



Human-in-the-loop

Before booking the workflow pauses. Approve resumes at the Booking Agent. Modify or Reject sends control back to the Planner. Booking runs only on the mock marketplace.

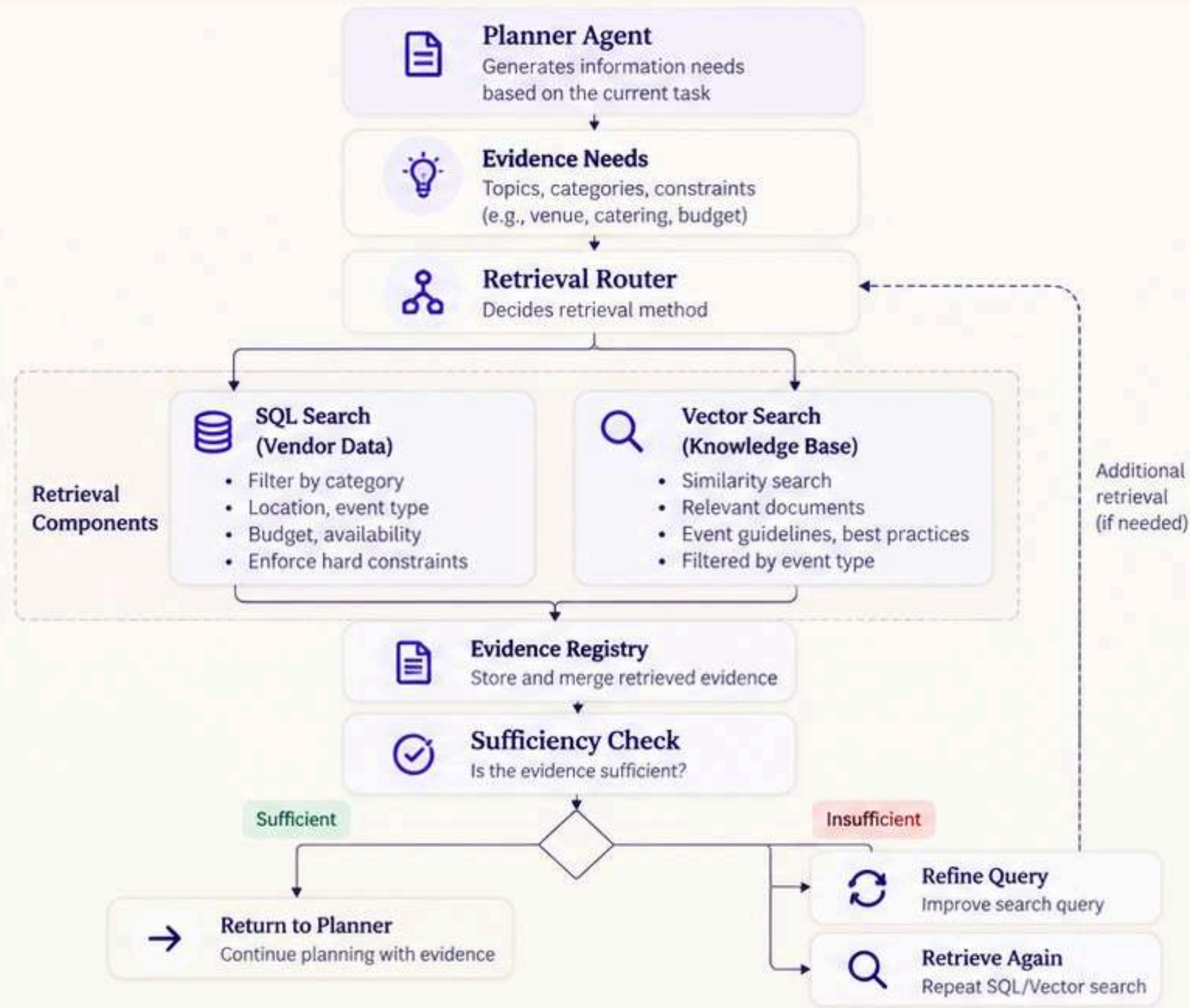


One shared state.
A self-correcting loop.
Human control at **key decisions**.



Agentic RAG Implementation

Adaptive retrieval with **evidence-aware decision making**.



How it works

1

Planner produces evidence needs

Based on the current planning task (e.g., venue, catering, budget).

2

Router selects SQL or vector search

Deterministic and dynamic routing based on the type of information needed.

3

SQL enforces hard constraints

Event type, category, city, budget, availability.

4

Vector search retrieves knowledge

Relevant documents and guidelines using pgvector, filtered by event type.

5

Evidence registry + sufficiency check

If the evidence is insufficient, the query is refined and retrieval is performed again.



Comparison of Retrieval Approaches



No-RAG

Prompt → LLM → Plan

- No external evidence
- Higher hallucination risk
- Limited constraint adherence



Basic-RAG

Fixed top-k retrieval → LLM

- Static retrieval
- Limited evidence coverage
- Better than no RAG but still incomplete



Agentic-RAG (Eventura)

Adaptive retrieval + sufficiency check

- ✓ Decides what evidence is needed
- ✓ Routes to SQL / vector sources
- ✓ Refines and re-retrieves if needed
- ✓ More grounded and constraint-aware planning



Implementation

Adaptive agentic retrieval with deterministic and dynamic routing, rather than fully autonomous LLM routing.



PostgreSQL
(Structured data)



pgvector
(Knowledge base)

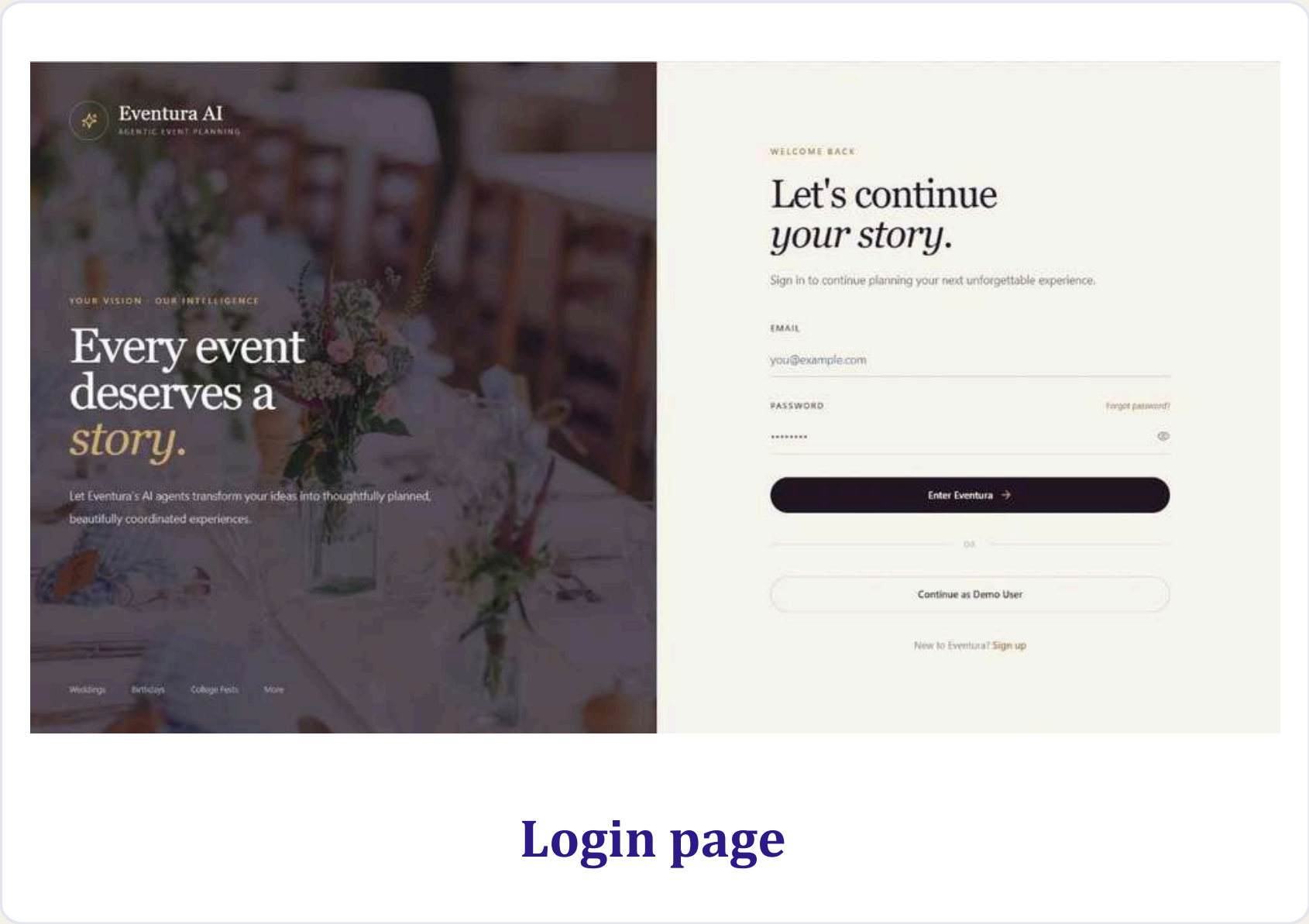


Evidence registry
(Merge & validate)



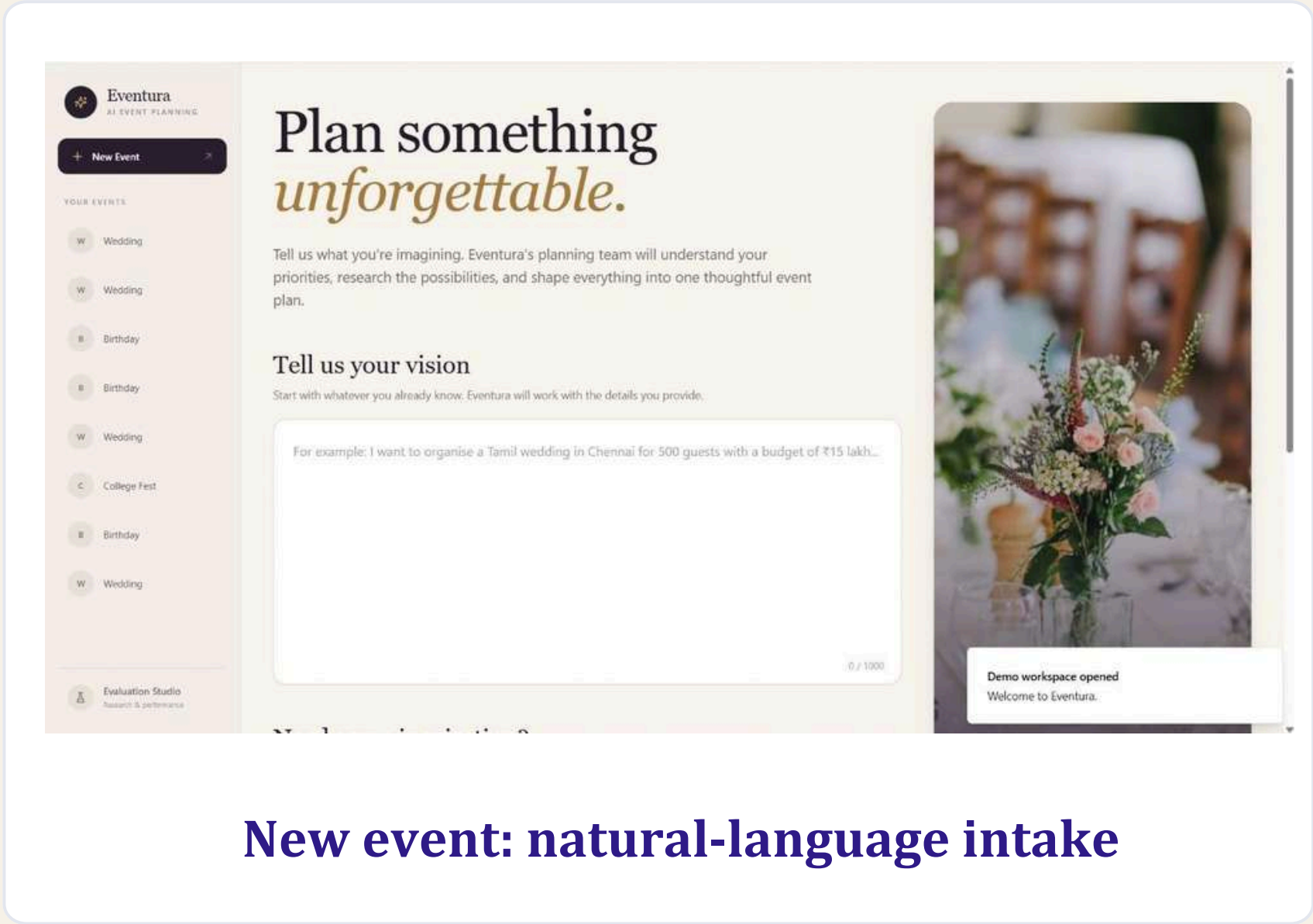
**RAG that knows
when it needs more evidence.**

Implementation: Login & New Event



The login page features a hero section on the left with a background image of a wedding table. It includes the Eventura AI logo, the tagline 'Every event deserves a story.', and a brief description of the AI event planning service. The right side contains a login form with fields for email and password, a 'Forgot password?' link, and buttons for 'Enter Eventura' and 'Continue as Demo User'. A link to 'Sign up' is at the bottom.

Login page



The 'New Event' page has a sidebar on the left with a 'New Event' button and a list of event categories (Wedding, Birthday, College Fest, etc.). The main area is titled 'Plan something unforgettable.' and includes a text input for 'Tell us your vision'. An example text is provided: 'For example: I want to organise a Tamil wedding in Chennai for 500 guests with a budget of ₹15 lakh...'. A 'Demo workspace opened' notification is visible at the bottom right.

New event: natural-language intake

Implementation: Dashboard & Plan Review

Eventura
AI EVENT PLANNING

+ New Event

YOUR EVENTS

W Wedding

W Wedding

B Birthday

B Birthday

W Wedding

C College Fest

B Birthday

W Wedding

Evaluation Studio
Research & performance

Wedding

EVENT WORKSPACE

Completed GEMINI AGENTIC BAG

EVENT PLAN

Wedding

20 December 2023 500 guests Chennai

View Full Plan Agent Activity Simulate Change

EVENT DATE

20 December 2023

GUEST COUNT

500 people

PLANNING BUDGET

₹15,00,000 50% allocated

EVENT DURATION

2 days

YOUR PLAN

Selected services

2 services selected

CATERING

Royal Catering Co. Chennai

Chennai

SELECTED PRICE

₹5,23,600

NEGOTIATED

₹71,400

PHOTOGRAPHY

Moments by Chennai

Chennai

SELECTED PRICE

₹1,56,740

NEGOTIATION

Savings found

QUALITY REVIEW

Plan needs refinement

FINANCIAL VIEW

Budget

50% ALLOCATED

BUDGET OVERVIEW

₹15,00,000

REMAINING

₹8,19,660

Event dashboard

Eventura
AI EVENT PLANNING

+ New Event

YOUR EVENTS

W Wedding

W Wedding

B Birthday

B Birthday

W Wedding

C College Fest

B Birthday

W Wedding

Evaluation Studio
Research & performance

Wedding

EVENT WORKSPACE

Completed GEMINI AGENTIC BAG

FINANCIAL PICTURE

Your budget at a glance

TOTAL BUDGET

₹15,00,000 Within budget

Allocated

₹7,51,740

₹7,51,740

REMAINING

₹7,48,260

NEGOTIATED SAVINGS

₹0

CURATED SELECTIONS

The team we've chosen

Select a vendor to mark it for replacement.

2 selections

CATERING

Royal Catering Co. Chennai

Chennai

FINAL PRICE

₹5,23,600

NEGOTIATED

Saved ₹71,400

PHOTOGRAPHY

Moments by Chennai

Chennai

FINAL PRICE

₹1,56,740

INDEPENDENT REVIEW

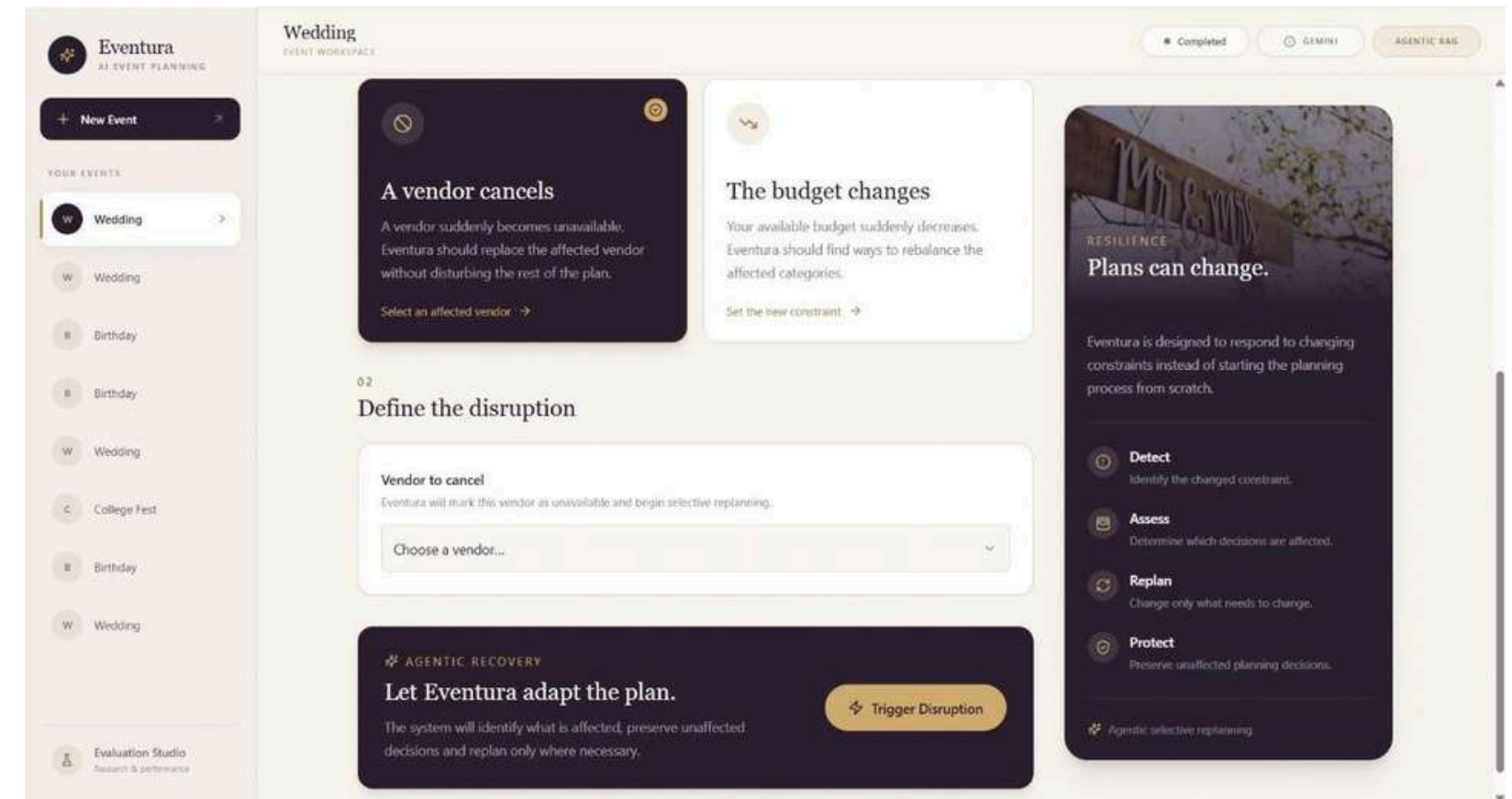
A second set of eyes

Final plan: budget and selected vendors

Implementation: Critic Review & Disruption



Critic feedback on the plan



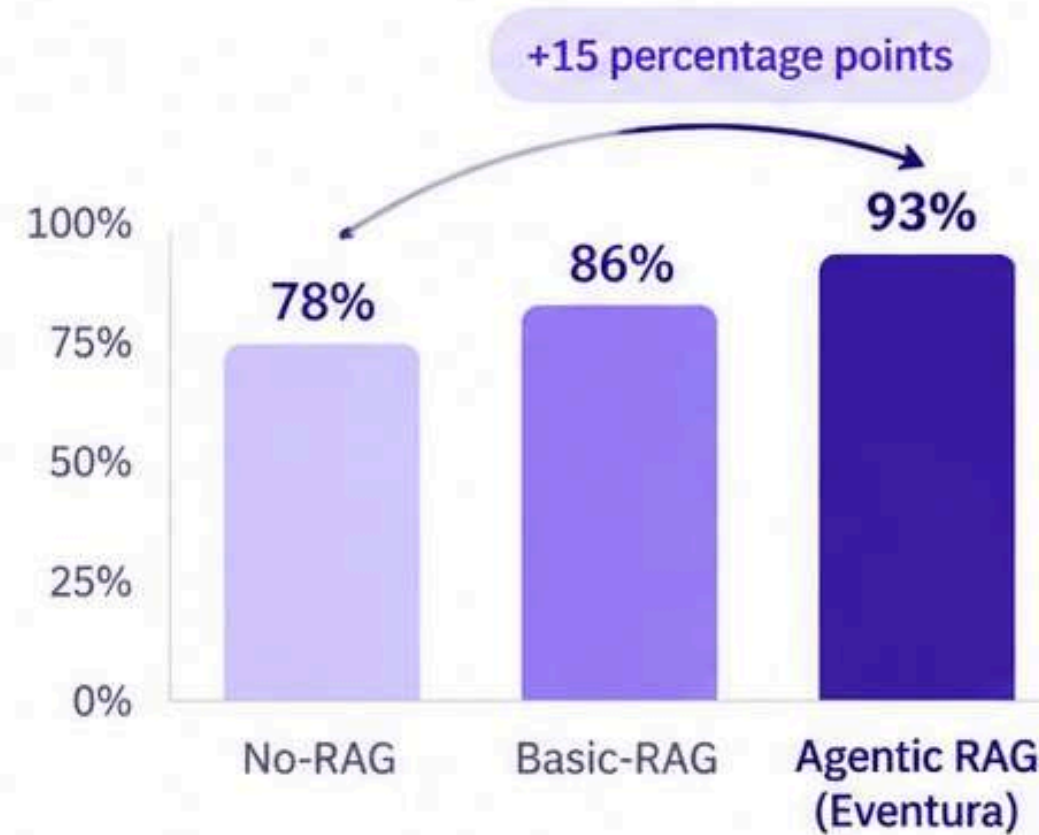
Disruption simulation and selective replanning

Conclusion & Future Work

Conclusion

Eventura treats event planning as a coordinated, evidence-grounded, human-approved workflow. Agentic RAG, the Critic loop and HITL approval raised constraint satisfaction from **78%** (No-RAG) to **93%** on the evaluated scenarios.

Constraint Satisfaction



Limitations



Synthetic vendors and mock booking



Limited to 3 event types



Limited knowledge base



Future Work



Real vendor APIs and live availability

Connect to real vendors and real-time inventory



Real payment and booking

Integrated payment flow with confirmation controls



Advanced negotiation

Packages, discounts, trade-offs and multi-vendor optimization



LLM-based retrieval controller

More autonomous and adaptive retrieval routing



Larger benchmark and continuous evaluation

More event scenarios and real-world datasets



Run-of-show, invitations and calendar sync

Generate schedules, send invitations and sync calendars



Multi-user collaborative planning

Shared planning, roles and permissions with cloud deployment



Thank You



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